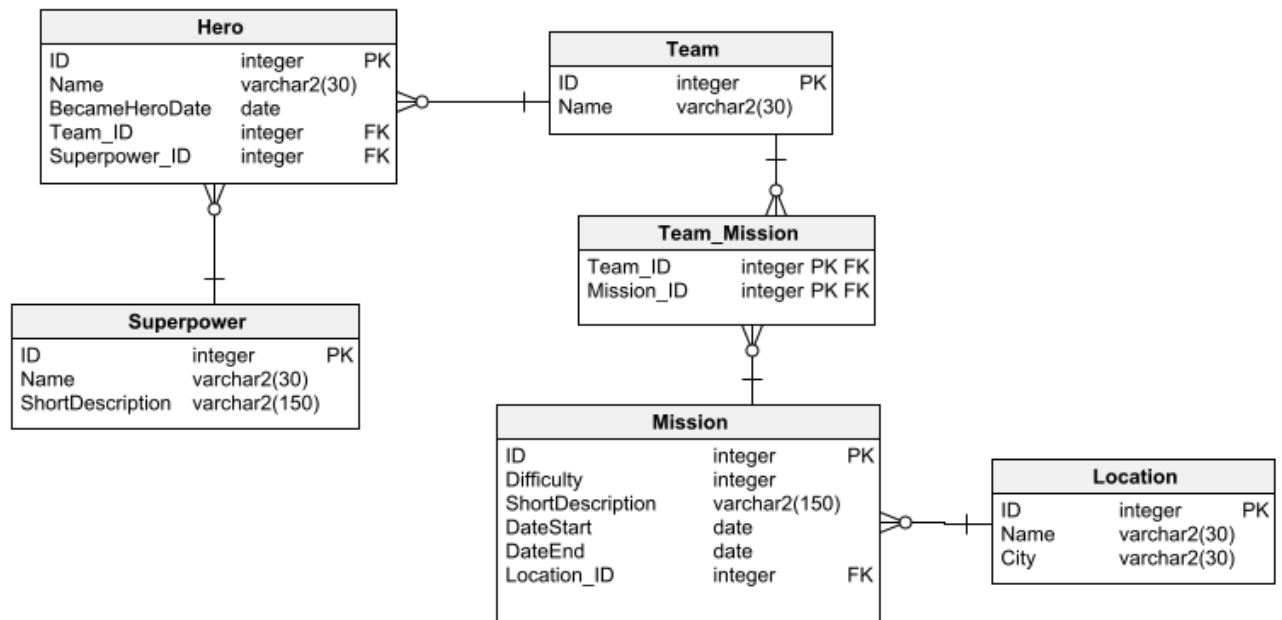


Example test



1)

Use a provided Heroes database to write the following queries. (Database code provided below)

You have to use the following structures at least once in one of the queries. You can receive 2 points per each structure:

- **Structure1**
- **Structure2**
- **Structure3**
- **Structure4**
- **Structure5**

1. Provide the hero who became a hero earliest in their team.
2. List the number of missions performed by each team. Do not list teams that have performed fewer than 2 missions.
3. Find the city where the most missions were executed.
4. List the duration of all missions. Include their description and the city location. Name the column with the time as "Mission Time".
5. Find the team that did not participate in any mission.

6. List the data as follows: hero name, name of their superpower, name of their team. Title the column with the superpower name as "Power" and the column with the team name as "Team". Sort the data in descending order by team name and ascending order by hero name.
7. List the missions with a difficulty greater than the minimum difficulty of missions performed in New York City.
8. Without using ORDER BY, list the top 3 most difficult missions. Sort them by difficulty.
9. List the number of missions completed by each hero. Do not list heroes whose name starts with the letter "B".
10. List the unique teams that participated in missions with both difficulty 1 and 4.

Database script:

```
-- tables
-- Table: Hero
CREATE TABLE Hero (
    ID integer NOT NULL,
    Name varchar2(30) NOT NULL,
    BecameHeroDate date NOT NULL,
    Team_ID integer NOT NULL,
    Superpower_ID integer NOT NULL,
    CONSTRAINT Hero_pk PRIMARY KEY (ID)
) ;

-- Table: Location
CREATE TABLE Location (
    ID integer NOT NULL,
    Name varchar2(30) NOT NULL,
    City varchar2(30) NOT NULL,
    CONSTRAINT Location_pk PRIMARY KEY (ID)
) ;

-- Table: Mission
CREATE TABLE Mission (
    ID integer NOT NULL,
    Difficulty integer NOT NULL,
    ShortDescription varchar2(150) NOT NULL,
    DateStart date NOT NULL,
    DateEnd date NOT NULL,
    Location_ID integer NOT NULL,
    CONSTRAINT Mission_pk PRIMARY KEY (ID)
) ;

-- Table: Superpower
```

```

CREATE TABLE Superpower (
    ID integer NOT NULL,
    Name varchar2(30) NOT NULL,
    ShortDescription varchar2(150) NOT NULL,
    CONSTRAINT Superpower_pk PRIMARY KEY (ID)
) ;

-- Table: Team
CREATE TABLE Team (
    ID integer NOT NULL,
    Name varchar2(30) NOT NULL,
    CONSTRAINT Team_pk PRIMARY KEY (ID)
) ;

-- Table: Team_Mission
CREATE TABLE Team_Mission (
    Team_ID integer NOT NULL,
    Mission_ID integer NOT NULL,
    CONSTRAINT Team_Mission_pk PRIMARY KEY (Team_ID,Mission_ID)
) ;

-- foreign keys
-- Reference: Hero_Superpower (table: Hero)
ALTER TABLE Hero ADD CONSTRAINT Hero_Superpower
    FOREIGN KEY (Superpower_ID)
    REFERENCES Superpower (ID);

-- Reference: Hero_Team (table: Hero)
ALTER TABLE Hero ADD CONSTRAINT Hero_Team
    FOREIGN KEY (Team_ID)
    REFERENCES Team (ID);

-- Reference: Mission_Location (table: Mission)
ALTER TABLE Mission ADD CONSTRAINT Mission_Location
    FOREIGN KEY (Location_ID)
    REFERENCES Location (ID);

-- Reference: Team_Mission_Mission (table: Team_Mission)
ALTER TABLE Team_Mission ADD CONSTRAINT Team_Mission_Mission
    FOREIGN KEY (Mission_ID)
    REFERENCES Mission (ID);

-- Reference: Team_Mission_Team (table: Team_Mission)
ALTER TABLE Team_Mission ADD CONSTRAINT Team_Mission_Team
    FOREIGN KEY (Team_ID)
    REFERENCES Team (ID);

-- data

-- Table: Superpower
INSERT INTO Superpower (ID, Name, ShortDescription) VALUES (1, 'Flight',
'Ability to fly');
INSERT INTO Superpower (ID, Name, ShortDescription) VALUES (2, 'Web-slinging',
'Ability to shoot webs and swing');
INSERT INTO Superpower (ID, Name, ShortDescription) VALUES (3, 'Lasso of
Truth', 'Compels truth from anyone bound by it');

```

```

INSERT INTO Superpower (ID, Name, ShortDescription) VALUES (4, 'Espionage',
'Skilled in undercover operations');
INSERT INTO Superpower (ID, Name, ShortDescription) VALUES (5, 'Godlike
strength', 'Possesses incredible strength');
INSERT INTO Superpower (ID, Name, ShortDescription) VALUES (6, 'Enhanced
Strength', 'Superhuman strength and endurance');
INSERT INTO Superpower (ID, Name, ShortDescription) VALUES (7, 'Gamma
Radiation', 'Ability to transform into a powerful green creature');
INSERT INTO Superpower (ID, Name, ShortDescription) VALUES (8, 'Vibranium
Suit', 'Enhanced senses and agility, superhuman strength');
INSERT INTO Superpower (ID, Name, ShortDescription) VALUES (9, 'Weather
Manipulation', 'Control over weather elements');
INSERT INTO Superpower (ID, Name, ShortDescription) VALUES (10, 'Ring of
Power', 'Ability to create constructs with the power of the ring');

-- Table: Team
INSERT INTO Team (ID, Name) VALUES (1, 'Justice League');
INSERT INTO Team (ID, Name) VALUES (2, 'Avengers');
INSERT INTO Team (ID, Name) VALUES (3, 'X-Men');
INSERT INTO Team (ID, Name) VALUES (4, 'Team X');

-- Table: Hero
INSERT INTO Hero (ID, Name, BecameHeroDate, Team_ID, Superpower_ID) VALUES (1,
'Superman', TO_DATE('2023-01-01', 'YYYY-MM-DD'), 1, 1);
INSERT INTO Hero (ID, Name, BecameHeroDate, Team_ID, Superpower_ID) VALUES (2,
'Spider-Man', TO_DATE('2023-02-15', 'YYYY-MM-DD'), 2, 2);
INSERT INTO Hero (ID, Name, BecameHeroDate, Team_ID, Superpower_ID) VALUES (3,
'Wonder Woman', TO_DATE('2023-03-10', 'YYYY-MM-DD'), 1, 3);
INSERT INTO Hero (ID, Name, BecameHeroDate, Team_ID, Superpower_ID) VALUES (4,
'Black Widow', TO_DATE('2023-04-20', 'YYYY-MM-DD'), 2, 4);
INSERT INTO Hero (ID, Name, BecameHeroDate, Team_ID, Superpower_ID) VALUES (5,
'Thor', TO_DATE('2023-05-05', 'YYYY-MM-DD'), 2, 5);
INSERT INTO Hero (ID, Name, BecameHeroDate, Team_ID, Superpower_ID) VALUES (6,
'Captain America', TO_DATE('2023-06-10', 'YYYY-MM-DD'), 2, 6);
INSERT INTO Hero (ID, Name, BecameHeroDate, Team_ID, Superpower_ID) VALUES (7,
'Hulk', TO_DATE('2023-07-20', 'YYYY-MM-DD'), 2, 7);
INSERT INTO Hero (ID, Name, BecameHeroDate, Team_ID, Superpower_ID) VALUES (8,
'Black Panther', TO_DATE('2023-08-05', 'YYYY-MM-DD'), 1, 8);
INSERT INTO Hero (ID, Name, BecameHeroDate, Team_ID, Superpower_ID) VALUES (9,
'Storm', TO_DATE('2023-09-15', 'YYYY-MM-DD'), 3, 9);
INSERT INTO Hero (ID, Name, BecameHeroDate, Team_ID, Superpower_ID) VALUES (10,
'Green Lantern', TO_DATE('2023-10-01', 'YYYY-MM-DD'), 1, 10);

-- Table: Location
INSERT INTO Location (ID, Name, City) VALUES (1, 'Secret Base', 'Metropolis');
INSERT INTO Location (ID, Name, City) VALUES (2, 'Avengers Tower', 'New York');
INSERT INTO Location (ID, Name, City) VALUES (3, 'Paradise Island',
'Themyscira');
INSERT INTO Location (ID, Name, City) VALUES (4, 'Main Square', 'Hollywood');

-- Table: Mission
INSERT INTO Mission (ID, Difficulty, ShortDescription, DateStart, DateEnd,
Location_ID) VALUES (1, 3, 'Rescue hostages from supervillain',
TO_DATE('2023-06-01', 'YYYY-MM-DD'), TO_DATE('2023-06-11', 'YYYY-MM-DD'), 1);
INSERT INTO Mission (ID, Difficulty, ShortDescription, DateStart, DateEnd,
Location_ID) VALUES (2, 2, 'Retrieve stolen artifact', TO_DATE('2023-07-10',
'YYYY-MM-DD'), TO_DATE('2023-07-23', 'YYYY-MM-DD'), 2);

```

```

INSERT INTO Mission (ID, Difficulty, ShortDescription, DateStart, DateEnd,
Location_ID) VALUES (3, 4, 'Defeat the alien invasion', TO_DATE('2023-08-15',
'YYYY-MM-DD'), TO_DATE('2023-08-19', 'YYYY-MM-DD'), 1);
INSERT INTO Mission (ID, Difficulty, ShortDescription, DateStart, DateEnd,
Location_ID) VALUES (4, 1, 'Investigate mysterious disappearances',
TO_DATE('2023-09-20', 'YYYY-MM-DD'), TO_DATE('2023-09-30', 'YYYY-MM-DD'), 3);
INSERT INTO Mission (ID, Difficulty, ShortDescription, DateStart, DateEnd,
Location_ID) VALUES (5, 2, 'Stop the bank robbery', TO_DATE('2023-10-05',
'YYYY-MM-DD'), TO_DATE('2023-10-15', 'YYYY-MM-DD'), 2);
INSERT INTO Mission (ID, Difficulty, ShortDescription, DateStart, DateEnd,
Location_ID) VALUES (6, 3, 'Retrieve stolen technology from a high-security
facility', TO_DATE('2023-11-19', 'YYYY-MM-DD'), TO_DATE('2023-11-21',
'YYYY-MM-DD'), 2);
INSERT INTO Mission (ID, Difficulty, ShortDescription, DateStart, DateEnd,
Location_ID) VALUES (7, 4, 'Stop the supervillain from destroying the city',
TO_DATE('2023-12-05', 'YYYY-MM-DD'), TO_DATE('2023-12-25', 'YYYY-MM-DD'), 1);
INSERT INTO Mission (ID, Difficulty, ShortDescription, DateStart, DateEnd,
Location_ID) VALUES (8, 2, 'Investigate paranormal activities in a haunted
mansion', TO_DATE('2024-01-15', 'YYYY-MM-DD'), TO_DATE('2024-01-17',
'YYYY-MM-DD'), 3);
INSERT INTO Mission (ID, Difficulty, ShortDescription, DateStart, DateEnd,
Location_ID) VALUES (9, 3, 'Rescue hostages from a terrorist organization',
TO_DATE('2024-02-20', 'YYYY-MM-DD'), TO_DATE('2024-02-25', 'YYYY-MM-DD'), 2);
INSERT INTO Mission (ID, Difficulty, ShortDescription, DateStart, DateEnd,
Location_ID) VALUES (10, 1, 'Defend the city against a giant monster attack',
TO_DATE('2024-03-05', 'YYYY-MM-DD'), TO_DATE('2024-03-25', 'YYYY-MM-DD'), 1);

-- Table: Team_Mission
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (1, 1);
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (1, 2);
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (2, 2);
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (2, 3);
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (3, 4);
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (3, 5);
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (1, 3);
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (1, 4);
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (2, 5);
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (2, 6);
INSERT INTO Team_Mission (Team_ID, Mission_ID) VALUES (3, 7);

```